BAZAARLENS PK

Pakistan Grocery Price Intelligence & City-Level Analysis

EXECUTIVE SUMMARY | Portfolio Data Analytics Project

350	17	5 weeks	11 items
Kaggle records	PBS cities	Weekly reports	Basket items

Project at a Glance

BazaarLens PK examines grocery price patterns across Pakistani cities using publicly available Kaggle and Pakistan Bureau of Statistics (PBS) data. It combines data understanding, cleaning, exploratory analysis, city comparisons, weekly price movement, and a fixed-basket comparison to turn raw price tables into practical insights.

Data & Approach

The project uses three data components: a Kaggle grocery dataset containing 350 records for 15 items across 5 cities (collected on 28 April 2025); a PBS monthly snapshot covering 51 items and 17 cities; and five PBS weekly reports from 1–29 October 2026, combined into 255 item-week records. Python, Pandas, NumPy, Matplotlib, Seaborn, and Plotly were used for preparation, analysis, and visualization.

Key Findings

Kaggle city comparison	Islamabad had the highest average recorded item price (Rs. 350.33), while Karachi had the lowest (Rs. 312.41).
Kaggle item comparison	Eggs (Dozen) had the highest average recorded price (Rs. 396.02); Moong Dal had the lowest (Rs. 243.84).
PBS monthly city comparison	Islamabad recorded the highest city average (Rs. 779.92) and Bannu the lowest (Rs. 674.84).
PBS weekly movement	Chicken farm broiler (live) increased by 18.02%, while wheat flour (20 kg) increased by 9.42% across the five weekly reports. Both peaked on 22 October before easing slightly by 29 October.
11-item grocery basket	The selected basket was most expensive in Islamabad (Rs. 8,226.68) and least expensive in Faisalabad (Rs. 7,164.49), a difference of Rs. 1,062.19.

Practical Implications

The results indicate meaningful differences in grocery prices across cities and products. A fixed basket can help consumers compare the cost of a consistent set of essentials, while weekly monitoring can highlight items with sharper short-term movements. This type of analysis may support price tracking, sourcing discussions, and consumer-facing comparisons.

Recommendations & Limitations

Recommendations

• Refresh weekly data regularly and track the same products over a longer period.

• Compare city prices using consistent item definitions and units wherever possible.

• Use the fixed basket as a repeatable benchmark and clearly state which products it includes.

• Add an interactive dashboard in a future version using Power BI or Tableau.

• Expand the analysis with more sources, cities, markets, and collection dates.

Important Limitations

• The Kaggle dataset is a single-day snapshot (28 April 2025), so it cannot show price trends over time.

• The PBS monthly dataset represents one monthly snapshot, not a long-term time series.

• The PBS weekly analysis covers five weeks, which is too short to establish long-term inflation trends.

• The 11-item basket is a selected comparison basket, not the complete cost of household groceries.

• Kaggle item prices may use different units (for example, eggs per dozen versus other products per kg/litre), so direct comparisons require care.

• City averages are analytical aggregates of available records and are not official CPI or cost-of-living measures.

Conclusion

BazaarLens PK demonstrates an end-to-end exploratory data analytics workflow using Pakistan-focused grocery price data. By combining multiple sources, comparing cities, tracking selected weekly changes, and evaluating a consistent grocery basket, the project provides a structured view of price differences and short-term movements. It can be expanded with more frequent data, broader coverage, and an interactive dashboard.

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